INCH-POUND

MIL-C-63513A (AR)
AMENDMENT 9
10 March 1995
SUPERSEDING
AMENDMENT 8
24 September 1992

MILITARY SPECIFICATION

CARTRIDGE, 105MM, TP-T, M490A1,
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-C-63513A (AR), dated 5 September 1985 and is approved for use by the U.S. Army Armament Research, Development and Engineering Center and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.2: Add the following under PRODUCT AND PACKAGING DRAWINGS:

"12935042 - Case, Cartridge, and Liner Assembly (SWCC)
12935040 - Cartridge, 105mm, TP-T, M490A1, (SWCC)".

PAGE 3

3.2: Delete in its entirety and substitute:

"3.2 Cartridges. The cartridges shall comply with all requirements specified on Drawing (dwg.) 9343009-1 or 9343009-2 or 12935040 (alternative) and with all requirements specified in applicable specifications."

3.3 Between last two words, "propellant" and "used", add: "and type of cartridge case".

3.5.1: Delete in its entirety and substitute:

"3.5.1 Muzzle velocity. The corrected mean muzzle velocity shall be not greater than 3885 feet per second (fps), (1184.15 meters per second (mps)) or less than 3815 fps (1162.81 mps) and the standard deviation for the lot shall be not greater than 9.5 fps."

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PAGE 4

Add paragraph 3.9:

"3.9 Cartridge case crimp (bullet pull). Drawing 9343009-1 or 9343009-2. The cartridge case shall separate from the projectile at a load between 4000 and 6000 pounds.

Drawing 12935040 - The cartridge case shall separate from the projectile at a load between 2300 and 6000 pounds."

PAGE 6

Drawing Number: Delete "9343009-1 or 9343009-2" and substitute "9343009-1 or 9343009-2/12935040".

PAGE 7

Drawing Number: Delete "9343009-1 or 9343009-2" and substitute "9343009-1 or 9343009-2/12935040".

PAGE 9

4.4.2.1:

Next Higher Assembly: Delete "9343010-1 or 9343010-2" and substitute "9343010-1 or 9343010-2/12935042".

Delete drawing number "9343011" and substitute "9343011 or 12934834".

Minor 202: Delete "Evidence of poor workmanship, including liner film torn or damaged" and substitute "Evidence of poor workmanship, including torn or damaged liner film, skirt or string".

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PAGE 10

4.4.2.2:

Delete: "203 End of cloth overlap 0.65% 3.2 Visual".

Drawing Number: Delete "9343010-1 or 9343010-2" and substitute "9343010-1 or 9343010-2/12935042".

Next Higher Assembly: Delete "9343009-1 or 9343009-2" and substitute "9343009-1 or 9343009-2/12935040".

Add Major 106:

"106 Evidence of contamination in the inside diameter of the cartridge case 100% 3.8 Visual".

Delete "Length of liner incorrect (Flush to 1/4 inch above mouth of case)" and substitute "Length of liner incorrect) Flush to 1/2 inch above mouth of case)."

PAGE 11

4.4.2.3:

Next Higher Assembly: Delete "9343009-1 or 9343009-2" and substitute "9343009-1 or 9343009-2/12935040".

Critical Defect: Add:

"2 Plug missing 100% 3.2 Visual".

PAGE 12

4.4.2.4:

Drawing Number: Delete "9343009-1 or 9343009-2" and substitute "9343009-1 or 9343009-2/12935040".

Delete: "201 Evidence of poor workmanship 1.65% 3.8 Visual" and substitute: "201 Evidence of poor workmanship 0.65% 3.8 Visual".

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PAGE 13

4.4.2.5:

Drawing Number: Delete "9343009-1 or 9343009-2" and substitute "9343009-1 or 9343009-2/12935040".

Major 102: Delete in its entirety and substitute the following:

"102 Liner flaps not folded over propellant and cemented or liner skirts not tightened and secured with draw strings using proper knot (Note 1) 100% 3.2 Visual".

Major 103: Add Note 2 to Examination or Test column.

Major 104: Delete in its entirety.

Add Note 2: "2. Only if the liner/adhesive method is utilized."

Major 102: Delete Examination or Test column in its entirety and substitute: "Liner flaps not folded over propellant and cemented or secured with draw strings".

Major 103: After examination or test item, add "(Note 1)".

Add Major 104: "104 Liner skirts not tightened over propellant with proper knot 100% 3.2 Visual".

Add Note 1: "1. Only one inspection is applicable depending on the method of closure utilized."

PAGE 14

4.4.2.6: Next Higher Assembly: Delete "9343009-1 or 9343009-2" and substitute "9343009-1 or 9343009-2/12935040".

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PAGE 15

4.4.2.7:

Drawing Number: Delete "9343009-1 or 9343009-2" and substitute "9343009-1 or 9343009-2/12935040".

Major 105: Delete in its entirety.

Delete Note 4 in its entirety.

Delete: "104 Gap between cartridge case and obturator, max. 0.40% 3.2 Gage" and substitute: "104 Gap between cartridge case and obturator, max. 100% 3.2 Gage".

Add Major 105: "105 Forward bourrelet diameter, max. (Note 4) 100% 3.1 Ring Gage".

Add Major 106: "106 Runout of cartridge assembly NA 3.2 Gage/4.5.5.1".

Add Note 4: "4. To be performed immediately prior to packaging."

PAGE 20

4.4.3.4.1 Delete in its entirety and substitute:

"Muzzle velocity. The lot shall be rejected if the standard deviation of the sample cartridges multiplied by the factor 0.70 exceeds the lot requirement."

PAGE 21

4.5.1.2: Add new paragraph 4.5.1.2 as follows:

"4.5.1.2 Contamination in the inside diameter Prior to assembly of liner to cartridge case, solvent wipe (Acetone, Spec O-A-5) inside diameter where the adhesive/liner touches the cartridge case. Solvent and wiper (lint free) shall be visually clean."

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4.5.5 Delete the entire paragraph and substitute the following:

"4.5.5 Profile and alignment check. The profile and alignment of each crimped cartridge shall be verified by a gage corresponding to drawing 8828098, which is to be horizontal ($0^{o} \pm 2.5^{o}$). The rear face of the cartridge shall be positioned within one inch of the gage's push arm and be inserted with a peak force not to exceed 60 pounds (27.2 kg) and a peak velocity of 2 ft/sec (61 cm/sec). No manual assistance (i.e., rotation, agitation, etc.) is permitted during inspection. Any cartridge which fails to be inserted or seated with the required force and velocity in any orientation and in accordance with the following sampling plan shall be rejected:

a. Inspect (50) cartridges in 4 angular orientations approximately 90° apart.

b. After (50) consecutive cartridges successfully chambergage, switch to chambergaging 1 out of 100 in 4 angular orientations and all others in 1 (one) random orientation.

c. Any failure will result in returning to step (a.).

When applicable, rejected cartridges which are reworked or repaired in accordance with Government approved procedures shall be reinspected in 4 angular orientations approximately 90° apart. Failure of cartridge to be inserted or seated in any orientation shall be cause for rejection of the cartridge. The profile and alignment gage shall not be utilized as an alignment tool."

Add the following paragraph:

"4.5.5.1 Runout inspection. Any cartridge that exceeds the runout dimension per drawing 9343009 shall be chambergaged in 4 angular orientations approximately 90 degrees apart (see 4.5.5). This feature shall be controlled using Statistical Process Controls in accordance with the Contractor's government approved plan."

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* 4.5.6.1.1: Delete in its entirety and substitute the following:

"4.5.6.1.1 Firing sequence. The data generated in this phase will primarily be used for determination of acceptance from a target impact dispersion and muzzle velocity standpoint and will require the firing of calibration rounds to establish a velocity correction. The firing sequence shall be five calibration rounds, fifteen test rounds and five calibration rounds. This phase shall be fired from a fixed-type facility mount with an M1 or M60A3 recoil utilizing a muzzle boresight or from an M1 or M60A3 tank with a gunner's auxiliary sight. An M68A1 gun tube shall be utilized as specified in Table I. All fixed type facility mounts and tank systems must be properly maintained and validated prior to testing."

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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